

MeriTayyari

Notes - Tuba

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Newton's Laws

physics - Grade 9th

$$F = ma$$

Newton's Laws (Full)

physics - Grade 9th

****Newton's Second Law of Motion****

- ****Statement****:

- The rate of change of momentum of an object is directly proportional to the net external ****force**** applied, and occurs in the direction of that force.

- ****Mathematical form (constant mass)****:

$$\mathbf{F}_{\text{net}} = m\mathbf{a}$$

- ****F**** = net external force

- ****m**** = mass of the body

- ****a**** = acceleration produced

- ****Key points****:

- Force, mass, and acceleration are vectors; direction matters.

- In absence of external force, an object maintains its state of motion (inertia).

- ****Applications****:

- Predicting motion of cars, rockets, or any accelerating body.

- Calculating required force to change velocity.